

AI Pseudonymizer for Medical Data

The Problem

Medical documents contain highly sensitive personally identifiable information, patient names, addresses, dates of birth, identification numbers, and clinical histories, that is strictly protected under GDPR and equivalent data protection frameworks. At the same time, modern AI-powered analysis tools offer significant potential value for clinical workflows, from summarizing complex patient histories to supporting diagnostic decisions and automating medical coding.

The core tension is that realizing this value typically requires sending documents to external AI providers, which creates an unacceptable privacy risk under current regulations. Existing approaches either forgo AI-assisted analysis entirely to preserve privacy, or transmit sensitive data externally without adequate protection. A principled, technically sound middle ground is largely absent from the tools available to medical facilities today.

This project addresses that gap by designing and implementing a local middleware system that allows medical documents to be analyzed by external AI providers without exposing patient identity — preserving both the utility of AI analysis and the integrity of patient privacy.

The Agentic Solution

A privacy-preserving middleware solution that enables secure, AI-powered analysis of sensitive medical documents. The system acts as a local intermediary, automatically pseudonymizing patient data in compliance with GDPR before transmitting documents to an external LLM provider for analysis. All pseudonymization and de-pseudonymization logic runs entirely on-device, ensuring that personally identifiable information never leaves the local environment unprotected. The system is designed for deployment within secure, controlled infrastructures such as medical facilities, where the environment itself provides the primary security guarantee.

Pseudonymization Approach

Rather than permanently storing identity mappings, the system maintains a temporary, session-scoped in-memory dictionary that maps each real PII token to a semantically plausible replacement for the duration of a document or batch session. Real names are replaced with invented but believable names, real addresses with fictional but realistic equivalents, and dates shifted consistently. This approach preserves the narrative flow and clinical semantics of the original document, ensuring the external LLM receives coherent, meaningful content, not redacted or garbled text.

Provider Agnosticism

The system is designed to be external LLM provider-agnostic, routing pseudonymized documents through a unified API gateway that supports any OpenAI-compatible endpoint. For the purposes of this student project, a free-tier provider such as Mistral or models available via OpenRouter is used to simulate a real-world external analysis service.

Analysis and Prompt Interface

The system offers a flexible prompting interface with a library of predefined templates for common medical analysis tasks, including Clinical Summarization, Diagnosis Support, and ICD Medical Coding. Users may also define custom prompts for ad-hoc analysis needs. The pipeline is explicitly designed to be extensible, allowing additional prompt templates or analysis tasks to be added with minimal effort.

Evaluation

A core deliverable of the project is a quantitative evaluation of pseudonymization fidelity, specifically the accuracy with which the system correctly reconstructs original PII tokens after the full analysis round-trip. This is measured using precision and recall against a ground-truth set of pseudonymized terms, providing a clear and objective success metric for the project.

Pipeline Components

The system is structured as a sequential multi-step pipeline, where each component handles a single, well-defined responsibility:

1. **Ingestion:** Receives the input document (PDF, text, DOCX, etc.) and converts it into a normalized plain text representation, passing clean text to the next stage.
2. **PII Detection:** Runs the normalized text through a local NLP model to identify and tag all PII entities such as names, addresses, dates, and ID numbers. Outputs an annotated version of the text with detected entities labelled by type.
3. **Pseudonymization:** Takes the annotated text and replaces each PII token with a semantically plausible substitute, building and maintaining the in-memory session dictionary (e.g. {"Hans Müller": "Erik Bauer", "Hauptstraße 5": "Gartenweg 12"}). Outputs the pseudonymized document and the dictionary.
4. **Analysis:** Takes the pseudonymized document and the selected prompt template or custom prompt, calls the external LLM, and returns the raw analysis result.
5. **De-pseudonymization:** Takes the analysis result and the session dictionary, and replaces all pseudonyms back with the original PII tokens. Outputs the final, restored analysis.

6. **Evaluation:** Compares the de-pseudonymized output against the original PII dictionary to compute precision and recall of correctly reconstructed tokens, flagging any terms that were missed or incorrectly restored.

Implementation Options

Default — Pure n8n: The entire pipeline is implemented as a self-hosted n8n workflow, where each component is represented as a dedicated node or sub-workflow. This approach prioritizes rapid development, visual transparency, and ease of demonstration.

Nice-to-Have — Hybrid n8n + LangGraph: For a more rigorous demonstration of agentic architecture, the PII Detection and Pseudonymization steps can be extracted and implemented as a dedicated LangGraph sub-agent, called by n8n via a local API endpoint. This combines the visual clarity of n8n with explicit, code-level agent reasoning.

Resources and Materials Required

Local LLM

Ollama used to run a small language model entirely on-device for PII detection and pseudonymization. Data never leaves the machine at this stage, which is the core privacy guarantee before any external communication occurs. Running a local Mistral model via Ollama is recommended, as it allows the entire system to use a single model family end-to-end.

External LLM Provider

Two free options are available, each with different trade-offs:

Option A — OpenRouter: A provider-agnostic API gateway supporting 200+ models through a single OpenAI-compatible endpoint. Students can switch between Qwen, Llama, Mistral, and others by simply changing a model string. Requires a free account, no credit card needed.

Option B — Mistral La Plateforme (Experiment Tier): The stronger choice from a GDPR perspective. Mistral operates entirely within European jurisdiction with servers in Paris, eliminating concerns around transatlantic data transfers. The Experiment tier provides free access to all Mistral models including Mistral Large and Small, with no credit card required — only phone verification. The rate limit of 2 requests per minute is sufficient for sequential document processing in a student project context. In a real-world deployment, regardless of provider choice, the external provider must be evaluated for GDPR compliance including data processing agreements and server location. The pseudonymization layer reduces but does not eliminate this obligation.

Dataset

A synthetic dataset of medical documents with embedded PII is generated using any available LLM. The model is prompted to produce realistic clinical notes for fictional patients, including all relevant PII fields — patient name, address, date of birth, physician name, and identification numbers. A minimum of nine documents should be generated: at least three suitable as input for each of the three prompt templates covering Clinical Summarization, Diagnosis Support, and ICD Medical Coding. These documents serve both as the primary test input and as the ground truth for the evaluation step.

Infrastructure

The system runs entirely within a self-hosted Docker environment, deployable on any standard machine within a secure local network such as a medical facility. No cloud infrastructure is required. n8n is self-hosted as the workflow orchestration layer, with all components including the local LLM, document storage, and the in-memory pseudonymization dictionary running within the same local environment.